

Joint Diffusion

$t = 0$

Forward Noising

$t = T$

Sequence

S_0

GGAGGAGUAACCGG

S_t

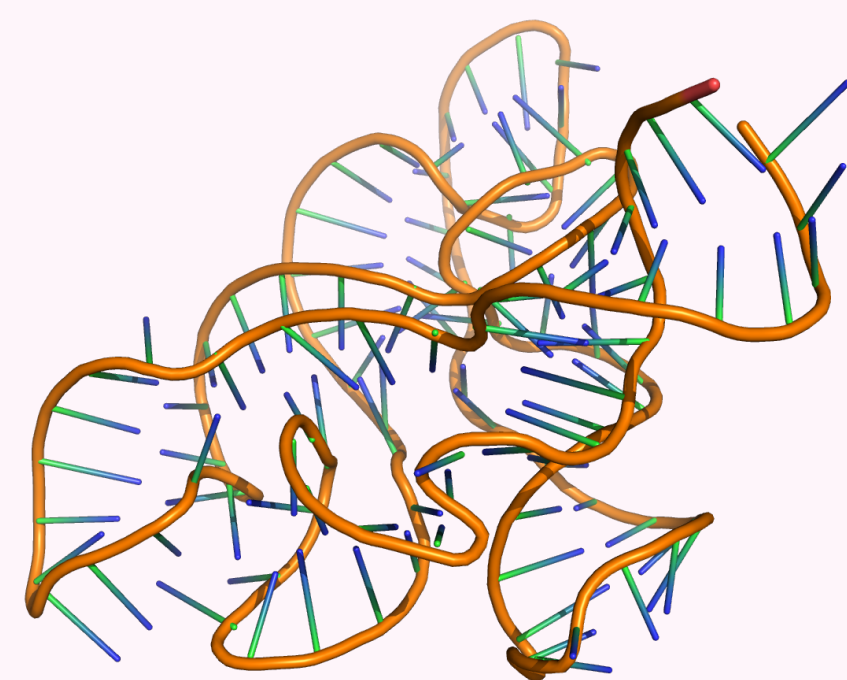
NNAGNANNAANNNG

S_T

NNNNNNNNNNNNNNNN

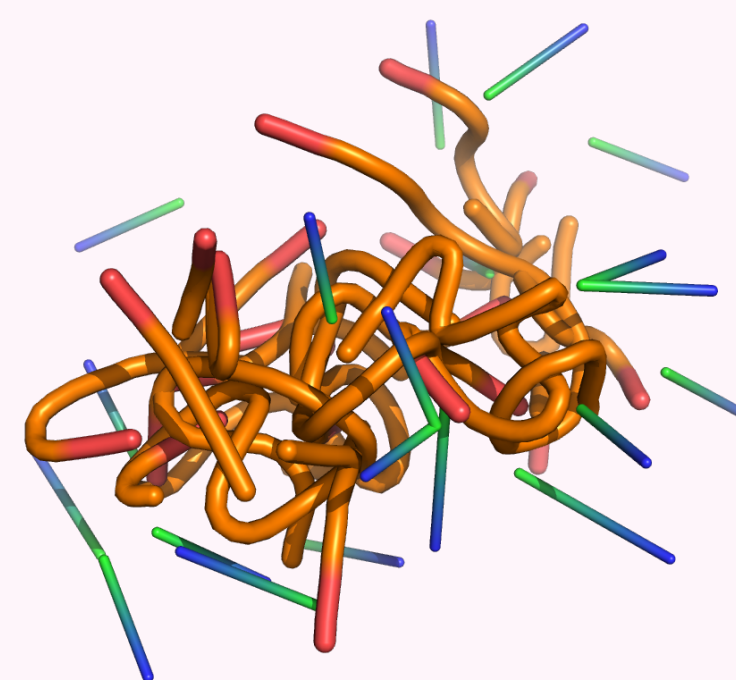
Discrete diffusion

Structure



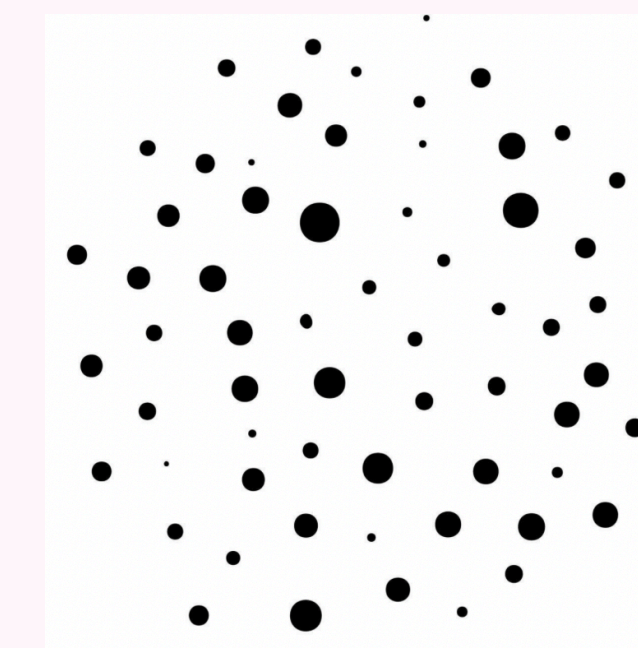
X_0

....

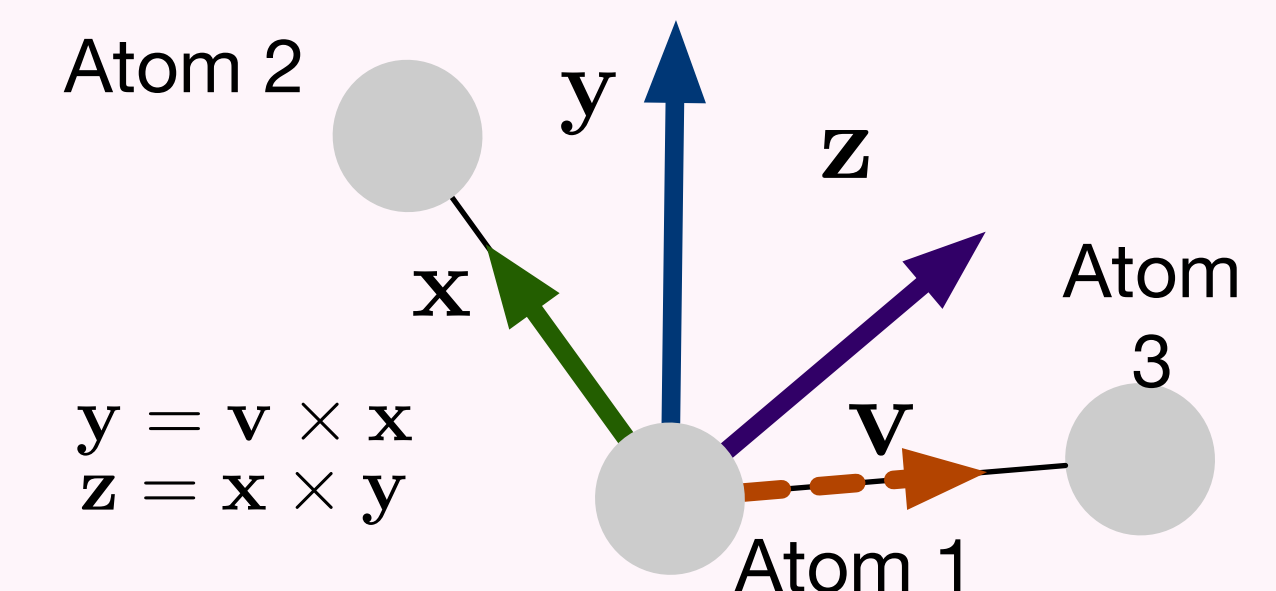


X_t

....



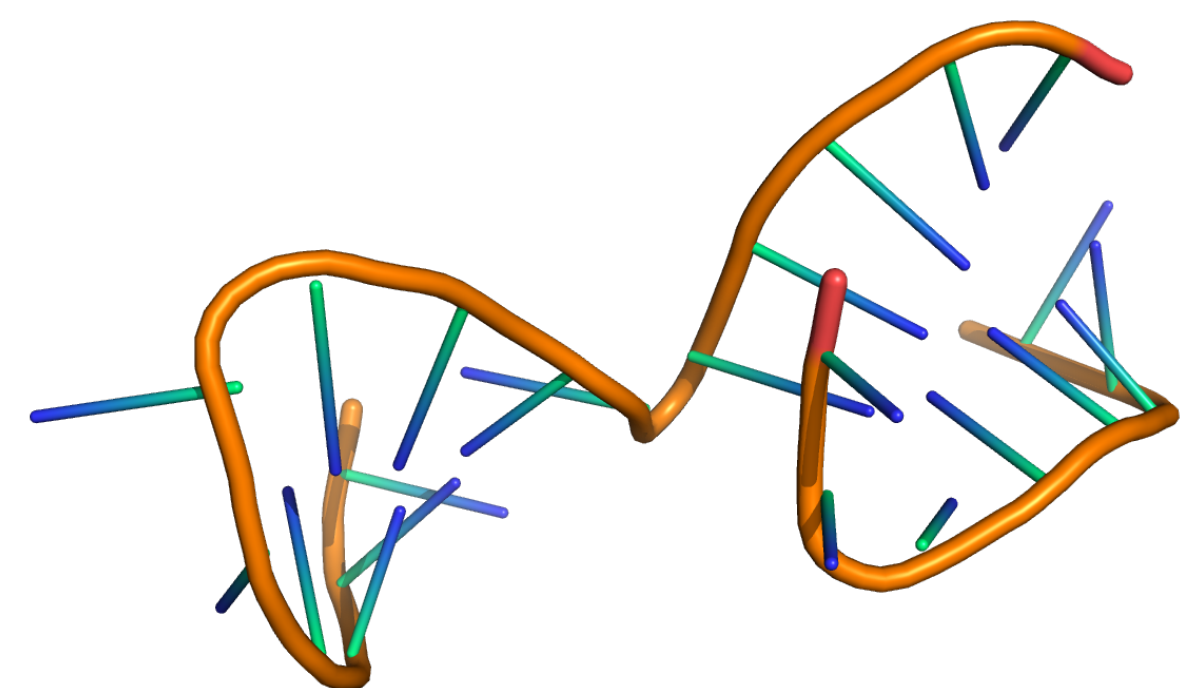
X_T



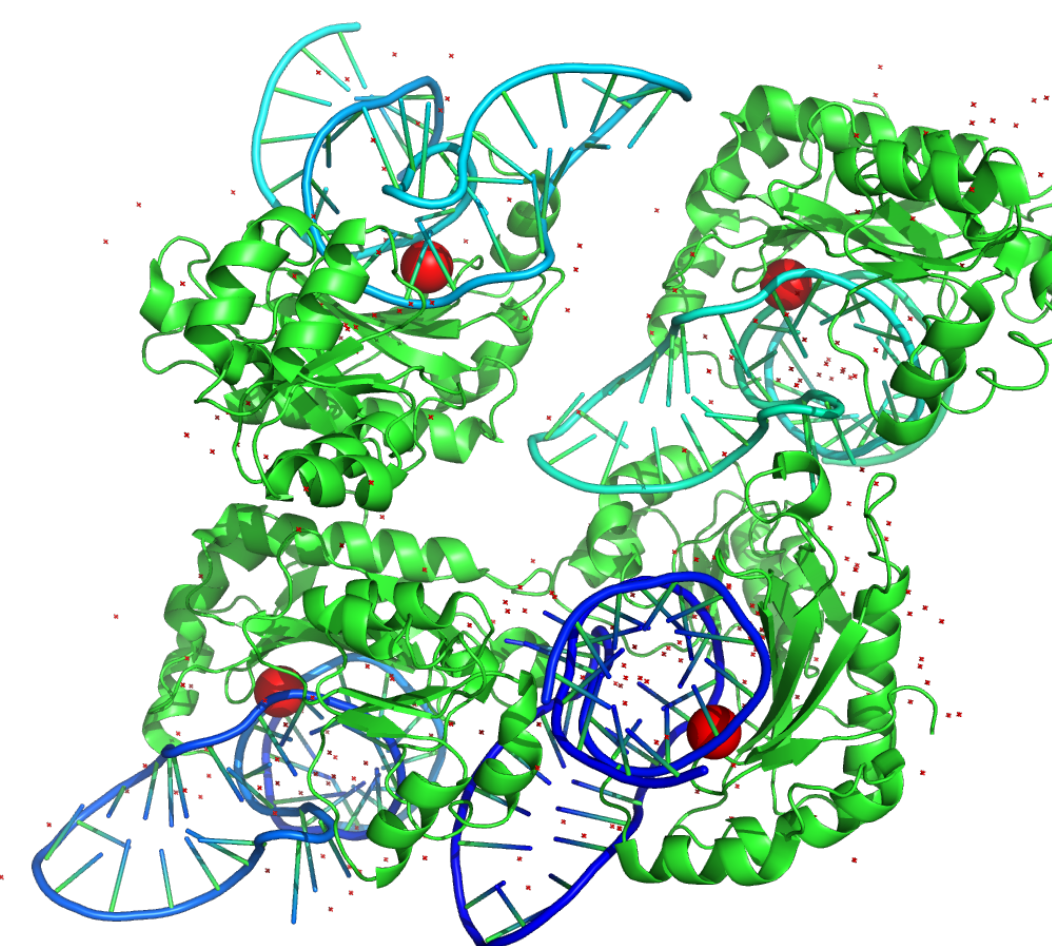
SE(3)-equivariant
continuous diffusion

$p_\theta(s_{t-1}, X_{t-1} | s_t, X_t)$

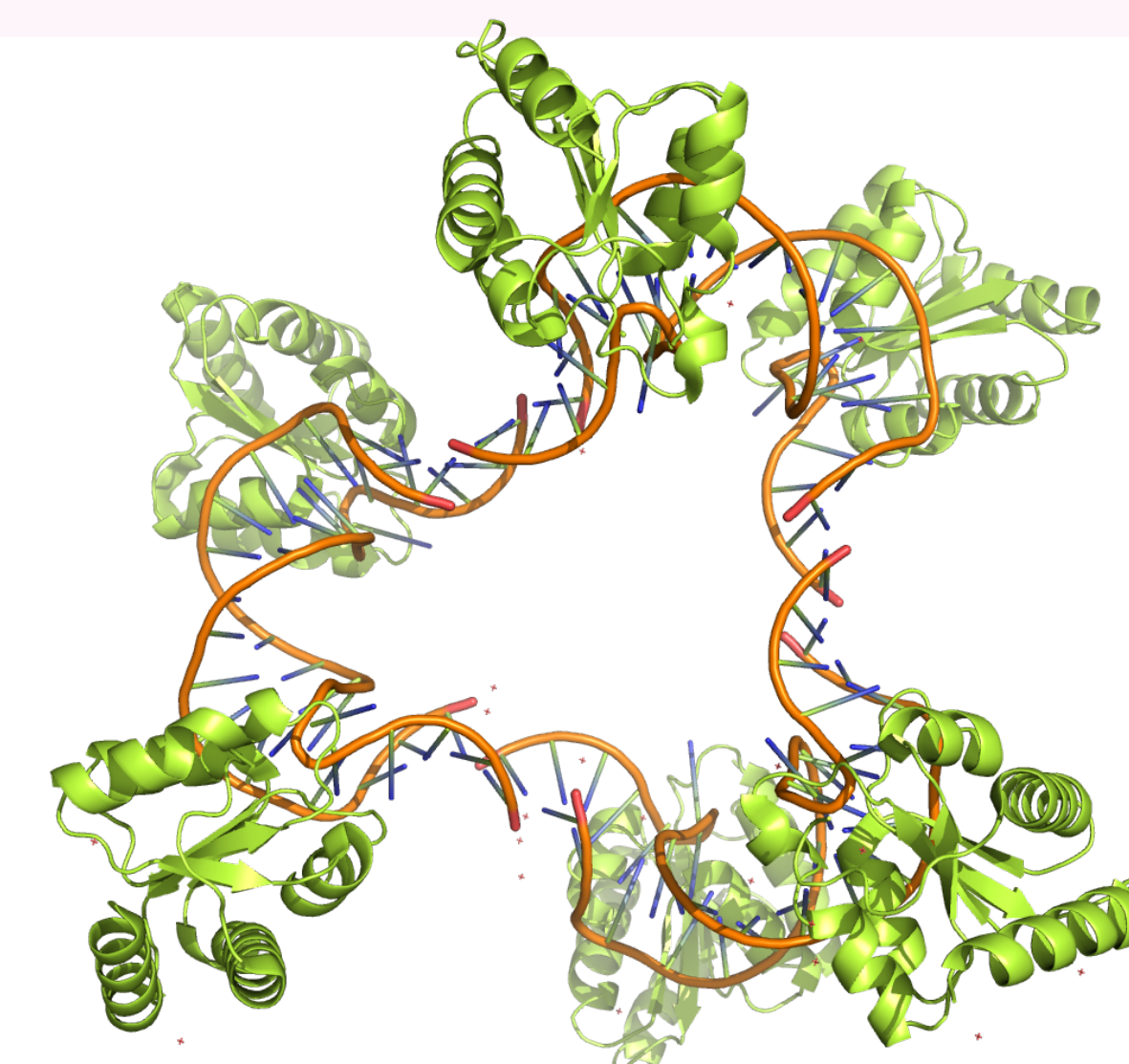
Fine-tuned RF2NA Denoiser



Solo RNA Design



Complex Design



Protein-conditioned RNA design